Homework 3

Brain-Computer Interaction, Fall 2026
Due Nov 11th

Overall goal

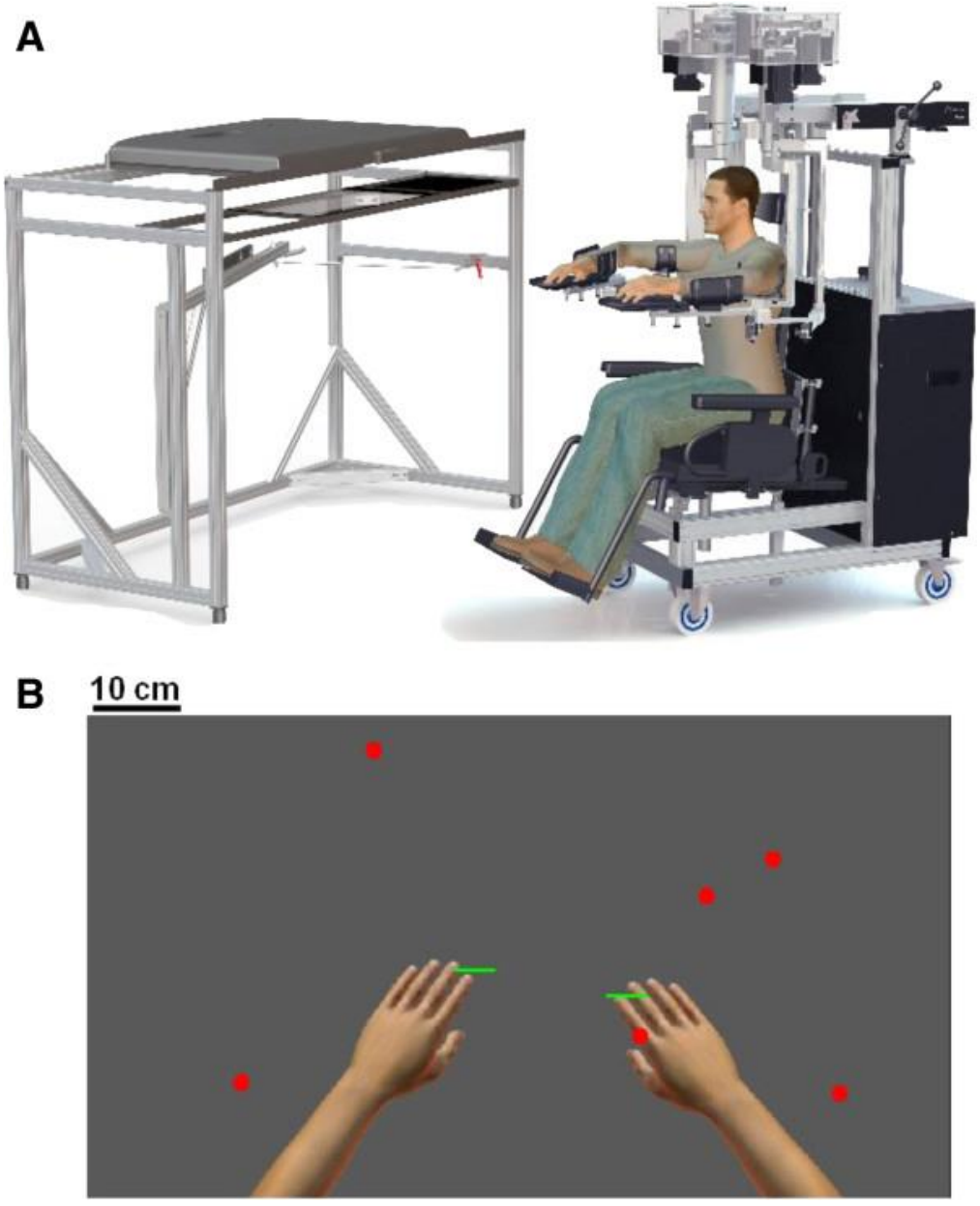
Classify neural data from an invasive recording to predict arm motion variables

Learn train-test splits, feature reduction, feature engineering

Background

Fagg, A. H., Ojakangas, G. W., Miller, L. E., & Hatsopoulos, N. G. (2009). **Kinetic trajectory decoding using motor cortical ensembles.**

- Implanted electrodes in monkey's arm area of motor cortex
- Two-joint robotic arm allowed free movement in horizontal plane
- Random-target pursuit task



Dataset

- 13000 samples
- **Input: neuronal activation patterns**
  - 48 neurons, firing rate within 50 ms time bin, 20 time bins representing 1 second delay
  - 960 total features
- **Output/prediction: arm motion**
  - Cartesian position (X) → position of hand in 2D space
  - Cartesian velocity (dX) → how fast and in what direction the hand is moving
  - Cartesian acceleration (ddX) → how quickly hand's speed and direction changing
  - Joint angle (theta) → angular position of shoulder and elbow joints
  - Joint angular velocity (dtheta) → how fast the angles are changing
  - Joing angular acceleration (ddtheta) → how fast the angular velocity of joints is changing
  - Torque → rotational force applied to each joint, represents muscle-generated effort

Methods

- Linear regression model: $y = X\theta$
- Cost: sum of squared error: $\min_{\theta} \|y - X\theta\|^2$
- Moore-Penrose pseudo-inverse method

$$\theta = (X^T X)^{-1} X^T y \quad \Rightarrow \quad \theta = X^+ y$$

- Fraction of Variance Accounted For (FVAF) $= 1 - \dfrac{\sum_{i=1}^{M} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{M} (y_i - \bar{y})^2}$
  - M is number of samples
  - numerator = residual variance
  - denominator = total variance of true data

Available functions

init

status

set_train([fold list])

set_test([fold list])

train

newwindow([figure#],[display type])

set_type('type')

set_pca([N dimensions])

set_proprioceptive(1)

set_delays(vec)

Train and Test splits

Training set → Used to *fit* the model (learn parameters)
Test set → Used to *evaluate* how well the model generalizes to unseen data

What is the caveat of having a small training set?

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The Curse of Dimensionality and Multicollinearity

- Ratio of number of features to number of samples
  - high (many samples, few features) → model can identify true patterns
  - low (many features, few samples) → model overfits noise/hallucinates
  - imagine trying to estimate someone's exam grade based on 100 different characteristics in only 3 students → many unique solutions
- Our case: features are n of neurons X time delays
  - 48 neurons x 20 bins = 960 features
  - 1100-1500 samples (within a training fold)



- Another issue: multicollinearity
  - features are highly correlated → unstable weights and overfitting
  - imagine solving a puzzle where puzzle pieces are identical → many unique solutions
- Mathematically, in both cases X^TX is ill-conditioned and non-invertible

Principal Component Analysis (PCA)

- Our data has many features, many of which are correlated/redundant
- PCA finds the main directions of variation
- PCA projects data onto new axes = components, where each component is a linear combination of original features
- Components are ranked: 1st components explains the most variance

What is the purpose of using PCA in our case?